

Experiment 14 — House Price Prediction

AIM

To implement a House Price Prediction model using an appropriate machine learning algorithm in Python.

SOFTWARE USED

- Python
- Google Colab / Jupyter Notebook
- Pandas
- NumPy
- Scikit-learn

PROCEDURE

1. Open Google Colab and create a new notebook.
2. Import the required Python libraries.
3. Load the house price dataset.
4. Display the dataset and inspect its features.
5. Select the input features and target variable.
6. Divide the dataset into training and testing sets.
7. Create a Linear Regression model.
8. Train the model using the training data.
9. Predict house prices using the test data.
10. Evaluate the model using Mean Squared Error (MSE) and R^2 Score.
11. Display the actual and predicted house prices.

CODE

```
import pandas as pd

from sklearn.datasets import fetch_california_housing

from sklearn.model_selection import train_test_split

from sklearn.linear_model import LinearRegression

from sklearn.metrics import mean_squared_error, r2_score

# Load California Housing dataset

housing = fetch_california_housing()

# Create DataFrame

df = pd.DataFrame(

    housing.data,

    columns=housing.feature_names

)

# Add target column

df["Price"] = housing.target

# Display dataset

print("Dataset Shape:", df.shape)

print("\nFirst 5 rows:")

print(df.head())

# Separate features and target

X = df.drop("Price", axis=1)

y = df["Price"]

# Split dataset

X_train, X_test, y_train, y_test = train_test_split(
```

```
X,  
y,  
test_size=0.2,  
random_state=42  
)  
# Create Linear Regression model  
model = LinearRegression()  
# Train model  
model.fit(X_train, y_train)  
# Predict house prices  
y_pred = model.predict(X_test)  
# Evaluate model  
mse = mean_squared_error(y_test, y_pred)  
r2 = r2_score(y_test, y_pred)  
print("\nMean Squared Error:", mse)  
print("R2 Score:", r2)  
# Display actual and predicted price  
print("\nActual Price:", y_test.iloc[0])  
print("Predicted Price:", y_pred[0])
```

OUTPUT

Dataset Shape: (20640, 9)

First 5 rows:

	MedInc	HouseAge	AveRooms	AveBedrms	Population	AveOccup	Latitude \
0	8.3252	41.0	6.984127	1.023810	322.0	2.555556	37.88
1	8.3014	21.0	6.238137	0.971880	2401.0	2.109842	37.86
2	7.2574	52.0	8.288136	1.073446	496.0	2.802260	37.85
3	5.6431	52.0	5.817352	1.073059	558.0	2.547945	37.85
4	3.8462	52.0	6.281853	1.081081	565.0	2.181467	37.85

	Longitude	Price
0	-122.23	4.526
1	-122.22	3.585
2	-122.24	3.521
3	-122.25	3.413
4	-122.25	3.422

Mean Squared Error: 0.5558915986952444

R2 Score: 0.5757877060324508

Actual Price: 0.477

Predicted Price: 0.7191228416019086

RESULT

Thus, the House Price Prediction Model was successfully implemented using the Linear Regression algorithm, and house prices were predicted based on the given housing features.